

Lucas Rodrigues

Full-stack Software Engineer | Divinópolis-MG, Brazil
lucasrodrigues.aes@gmail.com | linkedin.com/in/lucas-rodrigues

Professional Summary

Full-stack Software Engineer with 10+ years of experience delivering production-ready systems for industrial and commercial domains. Strong background in end-to-end development, including software architecture, backend and frontend engineering, system integrations, and scalable solutions for international clients, following clean and maintainable coding practices.

Professional Experience

Software Developer Consultant

TCS Industrial – Technologies & Creative Solutions | Belo Horizonte

Feb 2022 – Present

- Providing software development services to multinational industrial clients, including **Gerdau** and **Vallourec**.

Vallourec (2023 – Present)

- Maintaining and enhancing an industrial production traceability system.
- Developing communication drivers for integration with industrial equipment and production lines.
- Building internal projects mainly in .NET (C#), with React used on selected frontend initiatives.
- Designed and implemented a custom messaging system integrated with the traceability platform, delivering production events to a dedicated historical API as a scalable, decoupled NuGet package for multi-plant environments.

Gerdau (2022 - 2023)

- Modernized a legacy Delphi desktop system by redesigning the architecture around ASP.NET Core APIs.
- Implemented mathematical models for oxygen blowing and temperature control in converter furnaces.
- Developed an Angular frontend integrated with the new backend services.

Tech stack: .NET (C#), ASP.NET Core, PostgreSQL, Oracle, Angular, TypeScript, React, Windows Services, IPC,

Principal Software Engineer (Self-employed)

ARIA Engenharia de Software | Divinópolis

Aug 2020 – Present

- Delivered custom software solutions for clients across multiple industries.
- Designed, developed, and maintained production systems with real-world usage.

- Built and maintained internal frameworks, reusable components, shared libraries, and project boilerplates to standardize architecture and accelerate development.

Key Projects

Computer vision–based fruit classification system (production)

- Processing ~40 items per second in industrial environments.
- Built with .NET (C#), WPF, Windows Services, and TCP/IP socket communication.
- Integrated Python-based computer vision models and PostgreSQL.
- Deployed on three industrial machines in Brazil.

Nutrition monitoring SaaS platform (production)

- Designed and developed a B2C SaaS platform for nutritionists and patients.
- Implemented a PWA architecture using Vue 3 and Quasar.
- Integrated payments, authentication, and backend services using Firebase, Cloud Functions, and Node.js (Express).

Previous Project

- Developed a B2C food delivery platform for artisanal burger restaurants using Vue.js and GraphQL (Apollo Client), with a Node.js backend, Sequelize, and PostgreSQL.
- Integrated Google Maps, AWS S3, and Docker; project reached an advanced stage but was not launched due to market changes.

Tech stack: .NET (C#), WPF, Vue 3, Quasar, Vuetify, PostgreSQL, MongoDB, Firebase, Git Flow, Python, Computer Vision, gRPC, Node.js (Express)

CTO & Co-founder

FUSION Tecnologia | Divinópolis

Apr 2018 – Aug 2020

- Co-founded a software company delivering industrial and commercial solutions, including projects for multinational steel industry clients.
Defined technical strategy, development standards, and engineering best practices.
- Led the design and delivery of systems involving computer vision, web applications, and PWAs.
- Designed and implemented an industrial monitoring system for fine injection in blast furnaces using computer vision to support process control.
Co-founded **Checkbits**, a spin-off focused on digital checklists and system integrations for industrial and enterprise environments.
- Actively contributed to hands-on development and technical decision-making.

Tech stack: .NET (C#), Vue 3, Vuetify, PostgreSQL, Firebase, Git Flow, Python

Mechatronic Engineer Intern

Gerdau | Divinópolis

Apr 2016 – Apr 2018

- Worked in the Innovation and Technology team within steel manufacturing operations.
- Developed process optimization software to reduce operational costs and improve efficiency.

Tech stack: .NET (C#), Visual Basic, VBA, SQL Server

Education

Centro Federal de Educação Tecnológica de Minas Gerais (CEFET-MG – Divinópolis)
B.Sc. in Mechatronics Engineering (Robotics & Automation)

Jan 2011 – Dec 2015

Bayswater Language School | Cape Town, South Africa

English Language Immersion Program

B2+ English Proficiency Certification

Oct 2025

Technical Skills

Frontend

- JavaScript / TypeScript
- Vue 3, React, Angular
- Quasar, Vuetify, Material Design
- HTML5, CSS3
- PWA, UX/UI principles

Backend

- Node.js, Express
- ASP.NET (C#)
- Sequelize, Entity Framework
- Apollo Server, GraphQL

APIs & Data

- REST APIs, GraphQL
- Apollo Client
- SQL & NoSQL Databases (PostgreSQL, MSSQL, MongoDB, Firebase, Oracle)

Development Practices

- Git Flow
- Agile and Scrum-driven development
- Componentization, modular design, and reusable architectures

Additional Experience

- Desktop Applications: .NET Framework and .NET Core (C#, WPF, WinForms)
- Computer Vision: OpenCV (Python), EmguCV (C#)
- Advanced Excel - VBA